

Total No. of Questions : 4]

SEAT No. :

PA-10168

[Total No. of Pages : 1

[6010]-38

B.E. (Civil) (Insem)

GREEN STRUCTURES AND SMART CITIES

(2019 Pattern) (Semester - VIII) (Elective-VI) (401014E)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Figures to the right indicate full marks.
- 3) Draw neat figures wherever necessary.

- Q1)** a) Write in short on embodied energy. [4]
- b) Explain life cycle energy and its analysis. [6]
- c) Explain ecological footprints. [5]

OR

- Q2)** a) Write about bio capacity. [4]
- b) Explain embodied energy minimization concept and utilization for Civil Engineering material. [6]
- c) Explain operational energy in building. [5]

- Q3)** a) Write green construction practices. [4]
- b) Explain net zero building design. [6]
- c) Explain Energy Efficiency of Buildings. [5]

OR

- Q4)** a) Write Optimization Strategies in Buildings for a Sustainable Future. [4]
- b) Explain surface water balance. [6]
- c) Write the importance of rating and rating systems. [5]

❖ ❖ ❖